

Newman-Penrose Formalism Calculations

Stephani et al., Equation (28.27)

Metric

$$ds^2 = -2 H du \otimes du - du \otimes dr - dr \otimes du + \frac{r^2}{P^2} dx \otimes dx + \frac{r^2}{P^2} dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, r, x, y)$$

Metric Functions

$$H(u, r, x, y) = H$$

$$P(u, x, y) = P$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = (H) du + (1) dr$$

$$m = \left(\frac{\sqrt{2}r}{2P} \right) dx + \left(\frac{i\sqrt{2}r}{2P} \right) dy$$

$$\bar{m} = \left(\frac{\sqrt{2}r}{2P} \right) dx + \left(-\frac{i\sqrt{2}r}{2P} \right) dy$$

Contravariant components

$$l = (-1) \, dr$$

$$n = (-1) \, du + (H) \, dr$$

$$m = \left(\frac{\sqrt{2}P}{2r} \right) dx + \left(\frac{i\sqrt{2}P}{2r} \right) dy$$

$$\overline{m} = \left(\frac{\sqrt{2}P}{2r} \right) dx + \left(-\frac{i\sqrt{2}P}{2r} \right) dy$$

Directional Derivatives

$$D = -\partial_r$$

$$\Delta = -\partial_u + H \, \partial_r$$

$$\delta = \frac{\sqrt{2}P}{2r} \, \partial_x + \frac{i\sqrt{2}P}{2r} \, \partial_y$$

$$\bar{\delta} = \frac{\sqrt{2}P}{2r} \, \partial_x - \frac{i\sqrt{2}P}{2r} \, \partial_y$$

Spin Coefficients

$$\rho = \frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{H}{r} + \frac{\frac{\partial}{\partial u} P}{P}$$

$$\lambda = 0$$

$$\nu = \frac{\sqrt{2} P \frac{\partial}{\partial x} H}{2 r} - \frac{i \sqrt{2} P \frac{\partial}{\partial y} H}{2 r}$$

$$\alpha = \frac{\sqrt{2} \frac{\partial}{\partial x} P}{4 r} - \frac{i \sqrt{2} \frac{\partial}{\partial y} P}{4 r}$$

$$\beta = -\frac{\sqrt{2} \frac{\partial}{\partial x} P}{4 r} - \frac{i \sqrt{2} \frac{\partial}{\partial y} P}{4 r}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = -\frac{1}{2} \frac{\partial}{\partial r} H$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{\frac{\partial}{\partial x} P^2}{4r^2} + \frac{P \frac{\partial^2}{(\partial x)^2} P}{4r^2} - \frac{\frac{\partial}{\partial y} P^2}{4r^2} + \frac{P \frac{\partial^2}{(\partial y)^2} P}{4r^2} - \frac{H}{2r^2} - \frac{\frac{\partial}{\partial u} P}{2rP} + \frac{1}{4} \frac{\partial^2}{(\partial r)^2} H$$

$$\Phi_{12} = -\frac{\sqrt{2}P \frac{\partial^2}{\partial r \partial x} H}{4r} - \frac{i\sqrt{2}P \frac{\partial^2}{\partial r \partial y} H}{4r} - \frac{\sqrt{2} \frac{\partial^2}{\partial u \partial x} P}{4r} - \frac{i\sqrt{2} \frac{\partial^2}{\partial u \partial y} P}{4r} + \frac{\sqrt{2} \frac{\partial}{\partial u} P \frac{\partial}{\partial x} P}{4rP} + \frac{i\sqrt{2} \frac{\partial}{\partial u} P \frac{\partial}{\partial y} P}{4rP}$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = -\frac{\sqrt{2}P \frac{\partial^2}{\partial r \partial x} H}{4r} + \frac{i\sqrt{2}P \frac{\partial^2}{\partial r \partial y} H}{4r} - \frac{\sqrt{2} \frac{\partial^2}{\partial u \partial x} P}{4r} + \frac{i\sqrt{2} \frac{\partial^2}{\partial u \partial y} P}{4r} + \frac{\sqrt{2} \frac{\partial}{\partial u} P \frac{\partial}{\partial x} P}{4rP} - \frac{i\sqrt{2} \frac{\partial}{\partial u} P \frac{\partial}{\partial y} P}{4rP}$$

$$\Phi_{22} = \frac{P^2 \frac{\partial^2}{(\partial x)^2} H}{2r^2} + \frac{P^2 \frac{\partial^2}{(\partial y)^2} H}{2r^2} + \frac{\frac{\partial}{\partial r} H \frac{\partial}{\partial u} P}{P} + \frac{\frac{\partial}{\partial u} H}{r} - \frac{2H \frac{\partial}{\partial u} P}{rP} - \frac{2 \frac{\partial}{\partial u} P^2}{P^2} + \frac{\frac{\partial^2}{(\partial u)^2} P}{P}$$

$$\Lambda = -\frac{\frac{\partial}{\partial r} H}{3r} - \frac{\frac{\partial}{\partial x} P^2}{12r^2} + \frac{P \frac{\partial^2}{(\partial x)^2} P}{12r^2} - \frac{\frac{\partial}{\partial y} P^2}{12r^2} + \frac{P \frac{\partial^2}{(\partial y)^2} P}{12r^2} - \frac{H}{6r^2} - \frac{\frac{\partial}{\partial u} P}{2rP} - \frac{1}{12} \frac{\partial^2}{(\partial r)^2} H$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{\frac{\partial}{\partial r} H}{3r} + \frac{\frac{\partial}{\partial x} P^2}{6r^2} - \frac{P \frac{\partial^2}{(\partial x)^2} P}{6r^2} + \frac{\frac{\partial}{\partial y} P^2}{6r^2} - \frac{P \frac{\partial^2}{(\partial y)^2} P}{6r^2} + \frac{H}{3r^2} + \frac{1}{6} \frac{\partial^2}{(\partial r)^2} H$$

$$\Psi_3 = -\frac{\sqrt{2}P \frac{\partial^2}{\partial r \partial x} H}{4r} + \frac{i\sqrt{2}P \frac{\partial^2}{\partial r \partial y} H}{4r} + \frac{\sqrt{2}P \frac{\partial}{\partial x} H}{2r^2} - \frac{i\sqrt{2}P \frac{\partial}{\partial y} H}{2r^2} + \frac{\sqrt{2} \frac{\partial^2}{\partial u \partial x} P}{4r} - \frac{i\sqrt{2} \frac{\partial^2}{\partial u \partial y} P}{4r} - \frac{\sqrt{2} \frac{\partial}{\partial u} P \frac{\partial}{\partial x} P}{4rP} + \frac{i\sqrt{2} \frac{\partial}{\partial u} P \frac{\partial}{\partial y} P}{4rP}$$

$$\Psi_4 = \frac{P^2 \frac{\partial^2}{(\partial x)^2} H}{2r^2} - \frac{iP^2 \frac{\partial^2}{\partial x \partial y} H}{r^2} - \frac{P^2 \frac{\partial^2}{(\partial y)^2} H}{2r^2} + \frac{P \frac{\partial}{\partial x} H \frac{\partial}{\partial x} P}{r^2} - \frac{iP \frac{\partial}{\partial y} H \frac{\partial}{\partial x} P}{r^2} - \frac{iP \frac{\partial}{\partial x} H \frac{\partial}{\partial y} P}{r^2} - \frac{P \frac{\partial}{\partial y} H \frac{\partial}{\partial y} P}{r^2}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"